Revision highlights

Check if you can:

1. identify variable factors and fixed factors, variable costs and fixed costs;
2. define and differentiate the short run and the long run;
3. calculate total product, average product and marginal product;
4. state and illustrate the law of diminishing marginal returns;
5. calculate total cost, average cost and marginal cost;
6. explain the reasons for internal/external economies of scale and internal/external diseconomies of scale.

Key terms

variable factor 可變生產要素

variable cost 可變成本

fixed factor 固定生產要素

fixed cost 固定成本

short run 短期

long run 長期

total product 總產量

average product 平均產量

marginal product 邊際產量

law of diminishing marginal returns 邊際回報遞減定律

total cost 總成本

total variable cost 總可變成本

total fixed cost 總固定成本

average cost 平均成本

marginal cost 邊際成本

long run average cost 長期平均成本

economies of scale 規模經濟

diseconomies of scale 規模負經濟

optimal scale 最優規模

internal economies of scale 內部規模經濟

internal diseconomies of scale 內部規模負經濟

external economies of scale 外部規模經濟

external diseconomies of scale 外部規模負經濟

Exam-type questions

1. A supermarket plans to extend its opening hours to 24 hours. Which of the following are variable factors involved in the extra opening hours?

   (1) The premises
   (2) Full-time accounting clerk
   (3) Night shift security guard
   (4) Electricity
   (5) Delivery lorry
   (6) Part-time shop assistants

   A. (1), (2) and (5) only
   B. (3), (4) and (6) only
   C. (2), (3), (4) and (6) only
   D. (1), (3), (4), (5) and (6) only

2. John is a taxi owner who drives his own taxi. Which of the following are his fixed costs of providing taxi service?

(1) The taxi
(2) Annual licence fee of the taxi
(3) His human effort as a driver
(4) Fuel cost
(5) Insurance cost

A. (1) and (3) only
B. (2) and (5) only
C. (1), (2) and (5) only
D. (2), (4) and (5) only

3. The following table shows the input-output relationship of a firm which employs capital and labour only.

Capital (units)	Labour (units)	Marginal product (units)
2	1	6
2	2	7
2	3	5
2	4	2
2	5	-1

The average product of labour is the highest when the quantity of labour is _____ units, while the total product is the highest when the quantity of labour is _____ units.

A. 2 3
B. 2 4
C. 3 3
D. 3 4

4. The input-output relationship of a firm is shown in the table below. It only employs two factors, namely capital and labour.

Capital (units)	Labour (units)	Total output (units)
5	2	90
5	3	160
5	4	270
5	5	360

According to the table above, diminishing marginal returns set in when the _____ unit of labour is employed.

A. 2nd
B. 3rd
C. 4th
D. 5th

★**5.** The following table shows the average output of a firm. The only inputs required are labour and machines.

Machines (units)	Labour (units)	Average product (units)
2	2	13
2	3	14
2	4	15
2	5	16
2	6	X

If the law of diminishing marginal returns applies to the firm, the value of X

A. must be smaller than 16.
B. must be larger than 16.
C. can be any value greater than 16.
D. can be any value smaller than 16.

★**6.** The following table shows the production plan of a firm which only employs two factors, namely land and labour.

Land (units)	Labour (units)	Total product (units)
2	3	30
3	3	45
4	3	65
5	3	X
6	3	Y

Suppose the above data illustrate the law of diminishing marginal returns. Which of the following figures are consistent with the law?

(1) X = 65, Y = 45

(2) X = 90, Y > 110

(3) X > 93, Y = 121

A. (1) only

B. (2) only

C. (1) and (3) only

D. (2) and (3) only

7. When a firm only increases the amount of capital to increase its output,

A. labour productivity will decrease.

B. the optimal scale of production will decrease.

C. the average cost of production will eventually fall as a result of economies of scale.

D. the marginal product of capital will eventually decrease.

8. Which of the following statements about economies of scale is correct?

A. Only large firms enjoy economies of scale.

B. This is caused by the law of diminishing marginal returns in the long run.

C. It begins after a firm's output exceeds the optimal scale of production.

D. It can be the result of spreading the production cost over a larger output.

9. The following table shows the production data of a firm.

Output (units)	0	1	2	3	4
Total cost ($)	8	15	20	24	36

According to the above table, we can conclude that

A. the optimal scale of production is 3 units of output.

B. the firm enjoys economies of scale when it increases output from 2 units to 3 units.

C. The total variable cost is $12 when output is 2 units.

D. The average fixed cost does not vary with output.

10. The following table shows the production plan of a firm which only employs two factors, labour and capital.

Workers (units)	Average working hours per worker	Capital (units)	Average cost ($)
20	1	20	15
20	2	30	25
20	3	40	35
20	4	50	45
20	5	60	55

The above table shows a _____ situation in which _____.

A. short run ..... the law of diminishing marginal returns does not apply

B. short run ..... the law of diminishing marginal returns applies

C. long run ..... economies of scale exist

D. long run ..... diseconomies of scale exist

★**11.** The table below shows the total output of ABC ice cream shop per day. It employs only two inputs in production.

Premises (units)	2	2	2	2
Number of workers	10	10	10	10
Average working hours per worker	7	8	9	10
Total output (units)	3,400	3,700	3,900	4,000

As the market demand for ice cream increases, ABC ice cream shop raises its output by requiring workers to work longer hours, from 7 hours to 10 hours per day. Which of the following is correct?

A. The fixed cost is zero because the production cost of premises is zero.

B. The amount of labour employed by the firm remains unchanged.

C. The firm enjoys external economies of scale.

D. The change in output can be explained by the law of diminishing marginal product.

12. As the government implements policies to attract more IT companies to set up offices in the city, a small local IT company has to increase its salaries by 10% in order to hire graduates with an IT degree. The local company

A. enjoys internal economies of scale.

B. suffers internal diseconomies of scale.

C. enjoys external economies of scale.

D. suffers external diseconomies of scale.

★**13.** When a producer reduces all its inputs to cut its output by 50%, the total cost drops by 60%. Which of the following is a possible reason?

A. The law of diminishing marginal returns applies to the firm.

B. The organisation of the firm was too complicated before the reduction in output.

C. The interest rate of borrowing from banks increases after the reduction in output.

D. The firm can utilise its machines more fully after reducing its production scale.

14. A fast-food shop opens inside a theme park.

(a) Give TWO examples of variable costs incurred in the operation of the fast-food shop. Explain your answer. (3 marks)

(b) Give TWO examples of fixed costs incurred in the operation of the theme park. Explain your answer. (3 marks)

15. The input-output relationship of a firm is shown below. The only inputs required are labour and tools.

Labour (units)	Tools (units)	Total product (units)
20	2	50
21	3	55
22	4	62
23	5	70
24	6	76

State the law of diminishing marginal returns and explain whether the above data illustrate the law. (5 marks)

16. The following table shows the average product of a firm.

Labour (units)	Capital (units)	Average product (units)
10	2	200
10	3	250
10	4	350
10	5	500
10	6	595

Explain the economic principle illustrated by the data above. (5 marks)

17. Ken runs a bakery. Despite an influx of tourists during the summer, he is unable to hire more workers and can only purchase more tools to increase the daily production of bread.

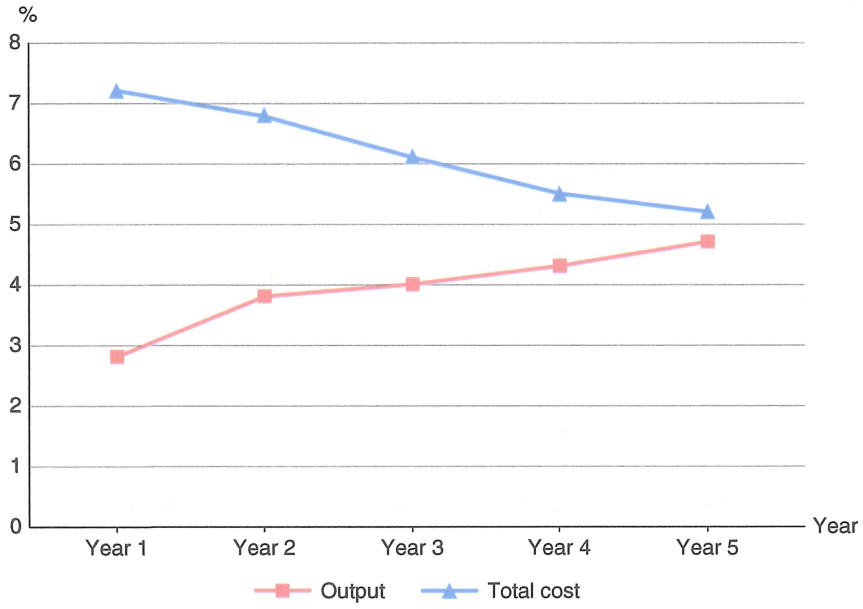
(a) Does Ken's bakery operate in the short run or long run? Explain your answer. (2 marks)

(b) After purchasing more tools, Ken found that the marginal product of tools has dropped. Which economic law can explain the above phenomenon? State the law and explain how it can be applied in this case. (6 marks)

18. After a bubble tea shop opened a new branch, the average cost of production decreased. Explain THREE possible reasons for such a change in average cost. (6 marks)

19. There are many cafés clustered in Sham Shui Po. In terms of average cost, explain TWO reasons for the phenomenon. (4 marks)

★**20.** DATA The CEO of an electric vehicle manufacturer implemented a plan to increase its production by expanding all inputs, including workers, factories and machines. The results of the plan are shown in the diagram below.



(a) Did the electric vehicle manufacturer operate in the short run or long run? Explain. (2 marks)

(b) Based on the above diagram, explain how the average cost has changed from Year 1 to Year 5. (2 marks)

(c) Explain the change in average cost with TWO possible economic reasons. (4 marks)

Extended activities

1. Students simulate running a shop (e.g., a lemonade stand). Assign roles like squeezing lemons, mixing lemonade and selling to customers. With a fixed amount of capital goods, start with a small team and gradually add more 'workers'. Track the total cups of lemonade sold and calculate the marginal product of each additional worker. Analyse how the marginal product changes as the number of workers increases.

2. Investigate how a local industry cluster (e.g., medical centres, toy shops, restaurants) benefits from external economies of scale. Identify specific benefits and explain how they reduce the firm's average cost.

3. Find a real-world example of a geographic region benefiting from external economies of scale (e.g., Silicon Valley). Describe how the concentration of firms in this area creates cost advantages.